

Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres with a CHIME FRB forecast

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids, $z < 0.114$, median $R_{\text{max}} = 11.9 h^{-1}\text{Mpc} = 17.0\text{Mpc}$, median $R_{\text{eff}} = 15.5 h^{-1}\text{Mpc} = 22.1\text{Mpc}$) with ZTF Bright Transient Survey supernovae using a Planck18-consistent KD-tree ball query with real SDSS mask. Of 6873 BTS transients at $z = 0.01\text{--}0.114$, 2465 lie in-mask: 183 void ($r/R_{\text{max}} < 0.8$), 271 shell ($0.8\text{--}1.0$), 1950 wall. Cosmology-grade (excluding 91T/91bg/pec/lax/CSM/SC): 2381 total, 176 void (141 normal Ia). Volume-limited $z \leq 0.06$: 1278 total, 71 void. Raw Poisson sensitivity to a void–wall Hubble offset is ~ 0.035 mag at 95 per cent including systematics floor; blinded SALT2 standardisation with mass, peculiar-velocity and selection controls is in progress. CHIME Catalog 1 (600 bursts, 524 sources) is used for footprint and error-budget demonstration only; unlocalised $z < 0.114$ stacking is underpowered by $> 10\times$ and joint constraints are deferred to Catalog 2. Catalog, randoms design and pipeline will be released.

Key words: large-scale structure of Universe – supernovae: general – fast radio bursts – methods: statistical

Table 1. Prior work vs this work.

Study	Voids	z	SNe	FRBs
Wang	lumped	< 0.05	hetero.	0 in voids
Tsaprazi	dens.+web	< 0.036	BTS+CLU	–
Aubert	DR7+Vor.	DR2 Ia	stretch	–
Sharma	BOSS VIDE	0.2–0.7	–	3455 stack
This work	VAST R_{max}	< 0.114	BTS+slice	Cat1 fcst

1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Table 1 summarises prior work. No published paper jointly uses VoidFinder DR7 spheres with flux-limited BTS and CHIME in the same low-redshift volumes; this work provides a supernova-main result with an FRB forecast in that geometry.

2 DATA

VAST Zenodo 7406035 v1.3.0; BTS Explorer quality and purity cuts (2026-09-30, 7143 rows); CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources: 506 one-off plus 94 bursts from 18 repeaters); NASA-Sloan Atlas v1_0_1 downloading for parent masses; Planck18 throughout.

3 METHODS

Void centres recomputed from RA, Dec, D ($D\text{ Mpc} = \text{Mpc}/h/0.7$) in the Planck18 frame; stored-XYZ residual median 0.00 Mpc. $R_v \equiv R_{\text{max}}$ core spheres; R_{eff} as systematics branch. Membership via ball query, $r/R_{\text{max}} = \min$, shell 0.8–1.0 excluded from primary. Mask from `NSA_main_mask.pickle` (360,180 bool); 1163/1163 void centres in-mask. Randoms: 200000 uniform comoving ($z = 0.01\text{--}0.114$) with seed 42, 35253 in-mask; expected $f_{\text{void}} = 0.174$, $f_{\text{shell}} = 0.150$, $f_{\text{wall}} = 0.677$; sky fraction 0.141, $V_{\text{eff,tot}} = 7.00 \times 10^7 \text{Mpc}^3$. Blinded SALT2 hierarchical Δ_{HR} with peculiar-velocity covariance, mass-step joint fit and BTS efficiency planned; this version reports mask-corrected counts with randoms volumes and a raw rate ratio.

4 RESULTS

Cuts flow: 7143 $\rightarrow$ 6873 (z) $\rightarrow$ 2465 (mask). Cosmology-grade 2381: void 176, wall 1883. Observed cosmo void fraction 176/2381 = 0.074 versus randoms 0.174: volume-corrected void/wall rate ratio 0.36 (Poisson rel. err 0.08 on void counts; BTS efficiency vs z not yet applied, so this is apparent, not intrinsic). Void Ia 141 normal ($4\times 91\text{T}$, $2\times \text{pec}$, $1\times \text{lax}$ excluded). Volume slice $z \leq 0.06$: 1278 total, 71 void. Threshold grid (cosmo): 0.6:59, 0.7:103, 0.8:176, 0.9:291, 1.0:438, 1.1:613. Redshift stability z_{min} 0.01:180, 0.02:175, 0.025:172, 0.03:168 void. Void median z 0.069 versus wall 0.057 confirms Malmquist gradient. FRB error budget (Cat1): per-sightline $\sigma_{\text{tot}} \sim 150\text{--}200 \text{pc cm}^{-3}$ (IGM 80–120, host 80–150, Milky Way residual 20–40 with NE2001–YMW16 median -3.3 , std 37.5, p90 68.1, halo 20). Void signal $\sim 10\text{--}30$. A $\delta\text{DM} < 50$ limit at 95 per cent needs $N_{\text{void}} \gtrsim 36$ (~ 200 localised total); unlocalised $z < 0.114$ needs ~ 900 void sightlines, have $\sim 30\text{--}50$.

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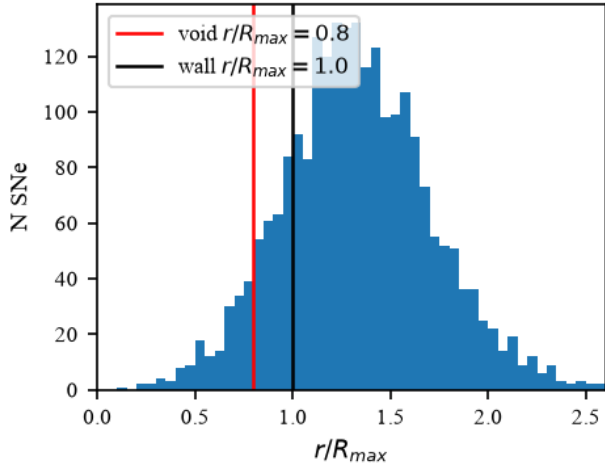


Figure 1. $r/R_{\max}$ distribution with 0.8 (void) and 1.0 (wall) lines.

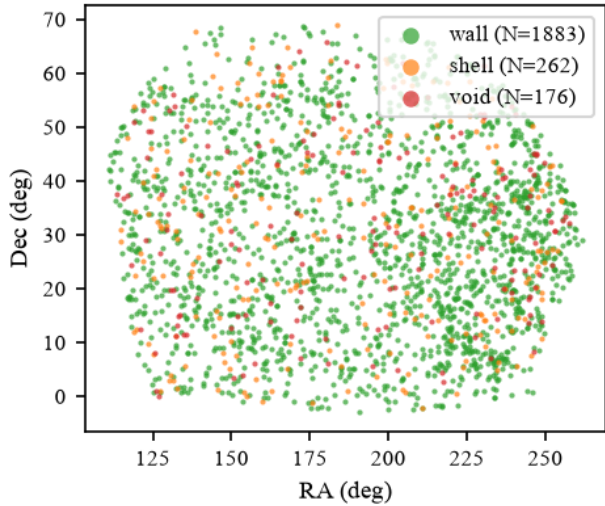


Figure 2. Masked RA–Dec distribution colour-coded by environment.

5 SYSTEMATICS AND NEXT STEPS

Peculiar velocities, selection efficiency, host masses, calibration, threshold grid for Hubble residuals, subtype cleaning done for counts, nulls (RA-shuffle within footprint, block bootstrap, 0.00/0.05 mag injection) planned. No unblinding until budget closed.

ACKNOWLEDGEMENTS

H. D. is a Higher Secondary Certificate student. This work used public archives only and received no institutional funding. The author thanks the VAST, ZTF, CHIME/FRB, SDSS, and NASA-Sloan Atlas teams for making their data and software public. AI agents (Muse Spark 1.3) assisted in building analysis codes, LaTeX files, and figures; however, the scientific analysis, interpretations, and conclusions remain the sole responsibility of the author.

6 DATA AVAILABILITY

VAST Zenodo DOI, BTS query URL and date with CSV hash, CHIME Vizier DOI, derived `b3_match_v2.parquet` with scripts tag v0.2 on GitHub on acceptance (reviewer token on request).

REFERENCES

- Douglass K. A. et al., 2023, *ApJS*, 265, 7
Perley D. A. et al., 2020, *ApJ*, 904, 35
Fremling C. et al., 2019, *arXiv:1910.12973*
CHIME/FRB Collab., 2021, *ApJS*, 257, 59
CHIME/FRB Collab., 2026, *arXiv:2601.09399*
Wang et al., 2026, *ApJ*, 997, 121
Sharma et al., 2026, *arXiv:2605.01994*
Tsaprazi et al., 2021, *MNRAS*
Aubert et al., 2025, *A&A*, 694, A7
Walker et al., 2024, *A&A*, 683, A71
Macquart et al., 2020, *Nature*, 581, 391